



Time-frequency analysis of pupil size modulated by global luminance, arousal, and saccade preparation signals using Hilbert-Huang transform

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ABSTRACT

Pupil size changes constantly and is mainly determined by global luminance signals. In addition, the pupil responds to various cognitive and arousal processes, with larger pupil dilation observed in higher levels of cognitive or arousal processing. Although these task-evoked pupillary responses are extensively used in the pupil research, pupil analysis focusing on the frequency domain, particularly in the context of arousal and cognitive modulations, is less established. Fourier Transform method (FFT) has been used to understand the modulation of task difficulty on pupil oscillations. However, physiological signals are often characterized as non-linear and non-stationary waves, and the conventional spectral analytical method with linearity presumption is less appropriate to reveal modulation dynamics between time and frequency. Here, we used Hilbert-Huang Transform (HHT) to examine the time-frequency modulations on pupil size regulated by arousal, cognitive, and global luminance signals. Consistent with previous research, using FFT, higher spectral densities were obtained with lower luminance background. Moreover, higher spectral densities were found in the high emotional arousal condition. With HHT, we further demonstrated temporal changes on amplitude spectrum and inter-trial phase coherence (ITPC) in each intrinsic mode function (IMF), with stronger amplitudes in higher IMFs (i.e., low frequencies). Moreover, although global luminance, arousal and saccade preparation modulated pupil oscillatory responses, the modulation pattern in different IMFs was different. Together, our results demonstrated dynamics between the time and frequency domain on pupil oscillatory responses, highlighting the importance of examining the time-frequency interactions in the context of various pupil modulations.

1. Introduction

Over the past decade, pupillometry has been greatly used in the fields of neuroscience and psychology, because pupil size being controlled by the sympathetic and parasympathetic system (May et al., 2019) is modulated by various cognitive and affective processes (Beatty, 1982; Bradley et al., 2008; Eckstein et al., 2017; Loewenfeld, 1999), in addition to its well-established modulation by global luminance level (Loewenfeld, 1999; McDougal and Gamlin, 2015). Converging evidence from human and animal studies has shown that pupillary responses correlate with neural activity of various brain structures (Aston-Jones and Cohen, 2005; Grueschow et al., 2021; Joshi et al., 2016; McGinley et al., 2015; Reimer et al., 2014, 2016; Varazzani et al., 2015; Yüzgeç et al., 2018), arguably mainly through the locus coeruleus-

norepinephrine system (Aston-Jones and Cohen, 2005; Mather et al., 2016; Sara, 2009; Sara and Bouret, 2012).

Seminal studies have shown that pupil size changes transiently in response to task events (Beatty, 1982; Bradley et al., 2008; Eckstein et al., 2017). These pupillary responses are commonly referred to as task-evoked pupillary responses, similar to an event-related brain potential that is time-locked to critical events for spontaneous electroencephalographic activity (Luck, 2014). A large body of studies have shown that larger task-evoked pupillary responses when the task demands a higher level of cognitive processing (Ayala and Niechwiej-Szwedo, 2021; Beatty, 1982; Dalmaso et al., 2020; van der Wel and van Steenbergen, 2018). For example, larger pupil dilation is observed in the more difficult condition in the short-term memory task (e.g., remembering more digits) (Kahneman and Beatty, 1966). Besides, task-

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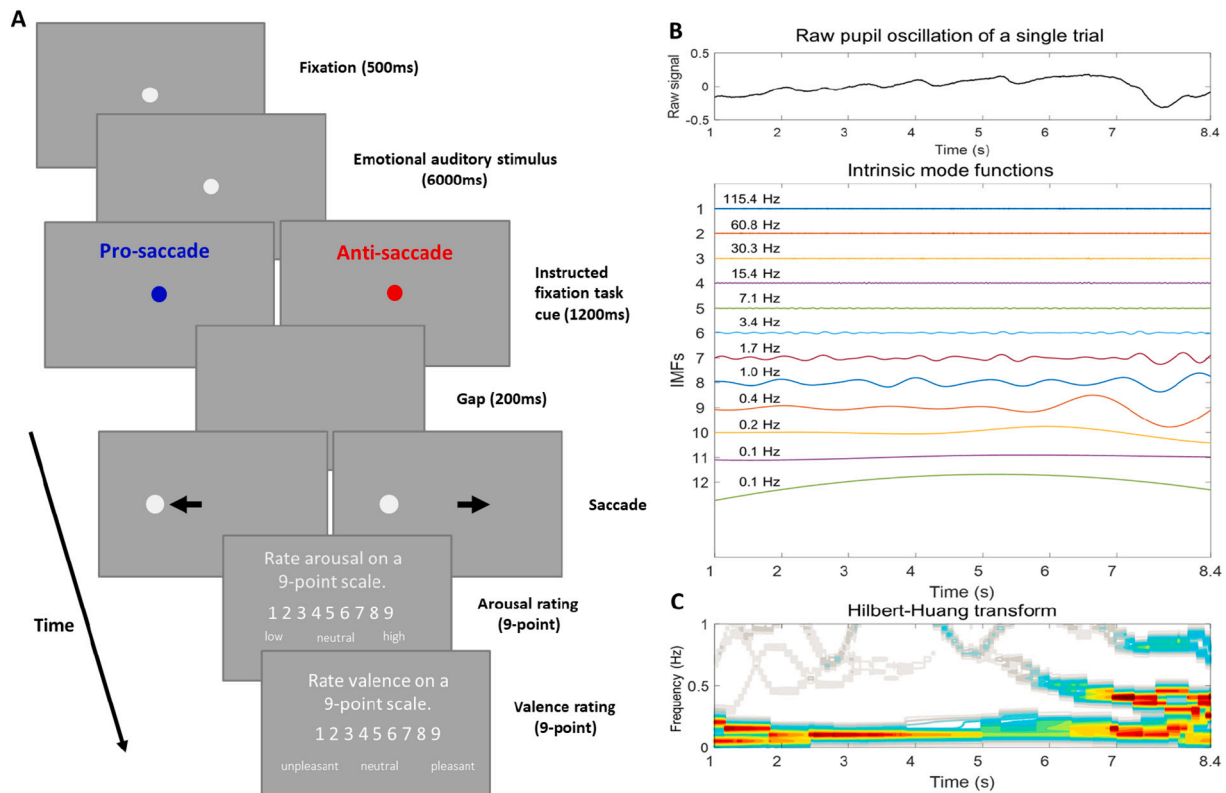


Fig. 1. (A) Experimental paradigm. Each trial began with a central fixation point on a background (Dark: 2 cd/m² or Bright: 16 cd/m²). After a delay, an emotional auditory stimulus was presented for 6 s, following with the instructed colored fixation cue (1200 ms) for pro- or anti-saccade condition. A blank screen was presented for 200 ms (gap) before target stimulus presentation, and participants were required to move their eyes to the target in the pro-saccade condition, or look away to the opposite location in the anti-saccade condition, after disappearance of the central fixation point. After that, participants were required to answer two questions about the sound presented. (B, C) Hilbert-Huang transform analysis of a single trial. Pupil dynamics after emotional sounds in a single trial in each IMF (B). The spectrogram of HHT obtained from all IMFs (C).

evoked pupillary responses are also modulated by arousal level (Bradley et al., 2008; Janisse, 1973), as pupil dilation is evoked by emotional stimuli and the magnitude of evoked dilation is modulated by emotional arousal level (Cherng et al., 2020; Janisse, 1973; Oliva and Anikin, 2018; Partala and Surakka, 2003). Although temporal analysis has been greatly used to pupil data (i.e., task-evoked pupillary responses), pupillary results focusing on the frequency domain is less established.

The spectral analysis has been used to examine the modulation of task difficulty on pupil size. Through fast Fourier transform (FFT), higher power in pupillary responses is generally obtained in the difficult condition (Murata and Iwase, 2002; Nakayama and Shimizu, 2004; Peysakhovich et al., 2015), though different effects are also noted (Murata and Iwase, 2002; Reiner and Gelfeld, 2014). Although the standard spectral analysis provides great insights into pupil oscillatory processing (Behrends et al., 2019; Lew et al., 2008; Lüdtke et al., 1998; Murata and Iwase, 2002; Naber et al., 2013; Nakayama and Shimizu, 2004; Nowak et al., 2008; Peysakhovich et al., 2015; Pomè et al., 2020; Reiner and Gelfeld, 2014; Ren et al., 2013; Schumann et al., 2020; Turnbull et al., 2017), the assumptions of linear and stationary signals may often be compromised, because physiological signals are characterized as non-linear and non-stationary waves (Boyina et al., 2012; Faust and Bairy, 2012; Zong and Chetouani, 2009).

Hilbert-Huang Transform (HHT) has been developed to analyze non-linear and non-stationary data, and is greatly applied to many areas of research (Huang, 2014; Huang et al., 1998). Basically, there are two major processes involved in the HHT, namely, the empirical mode decomposition (EMD) and the Hilbert spectral analysis (HSA). EMD, as an adaptive and data-driven manner, decomposes the intrinsic nature of the raw signal adaptively into single modes without the linearity assumption, known as intrinsic mode functions (IMFs). Specifically,

each successive EMD generates an IMF with lower oscillations than the previous IMF and naturally retains the physical properties of the signal. After decomposition, Hilbert transform can then be applied to any IMF to gather the local energy and instantaneous frequency information to produce HHT. Thus, HHT could overcome the limitation of the existing spectral analytical methods (e.g., Fourier and Wavelet methods) to investigate dynamic changes in both time and frequency domain in the pupil data. Although a few studies have used this method to analyze pupil data (Villalobos-Castaldi et al., 2016, 2017), research has been focused on the spontaneous oscillation. How do emotional and cognitive signals modulate time-frequency pupillary responses remains largely unknown.

The goal of this study is to use HHT to examine the time-frequency modulations on pupil size induced by arousal, cognitive, and global luminance signals. Emotional stimuli and saccade preparation cues are used because they can induce arousal and cognitive processes, respectively (Bradley et al., 2008; Munoz and Everling, 2004). We hypothesize that pupil oscillatory responses in the time and frequency domain could be modulated differently by luminance, arousal, and saccade preparation signals. Thus, effects of luminance, arousal, and saccade preparation could be more pronounced in a different time windows with specific frequency ranges.

2. Materials and methods

2.1. Experimental setup

Aspects related to task-evoked pupillary responses have been published, and the general method related to the task and participants are described in detail previously (Cherng et al., 2020), here we focused on

the time-frequency analysis. All experimental procedures were reviewed and approved by the Institutional Review Board of Taipei Medical University, Taiwan, and were in accordance with the Declaration of Helsinki (World Medical Association, 2001). Twenty-six participants (mean age: 25.7, SD: 4 years, 13 males) were recruited via an advertisement posted on the Taipei Medical University website. Participants had normal or corrected-to-normal vision and were naïve regarding the purpose of the experiment. Participants provided informed consent and were compensated financially for their participation.

2.2. Recording and apparatus

Participants were seated in a dark room. Eye position and pupil size were measured with a video-based eye tracker (Eyelink-1000 plus binocular-arm, SR Research, ON, Canada) at a rate of 500 Hz with binocular recording (left pupil was used). Stimulus presentation and data acquisition were controlled by Eyelink Experiment Builder and Eyelink software. Auditory stimuli were presented using headphones. Visual stimuli were presented on an LCD monitor at a screen resolution of 1920×1080 pixels (60 Hz refresh rate), subtending a viewing angle of $58^\circ \times 32^\circ$, with the distance from the eyes to the monitor set at 60 cm.

2.3. Emotional sound with pro- and anti-saccade task (Fig. 1A)

As described previously (Cherng et al., 2020), each trial began with the appearance of a central fixation point (FP) (0.5° diameter, gray color with 20 cd/m^2) on a background (Dark: 2 cd/m^2 or Bright: 16 cd/m^2). After 500 ms of the FP, an emotional auditory stimulus from the International Affective Digital Sounds (Bradley and Lang, 2007) was presented for 6000 ms. Afterward, the color of the FP changed in accordance with the saccade task condition (two FP colors, 20 cd/m^2). In pro-saccade trials, participants were instructed to look towards the peripheral target stimulus as soon as the target appeared. In anti-saccade trials, participants were instructed to look in the mirror location of the target stimulus as soon as the target appeared. After another 1200 ms of central fixation, the FP disappeared for 200 ms (gap) before the peripheral target stimulus appeared (0.5° diameter; yellowish dot with luminance 270 cd/m^2) to the left or right of the FP (8° eccentricity on the horizontal axis). This was followed by two questions presented on the screen. The participants were asked to rate the degree of arousal and valence of the stimuli on nine-point scales and using the keypad to respond (Libby et al., 1973; Wang et al., 2018). The participants came in twice (one week apart) for the same experiment with a different background luminance level (background luminance order counterbalanced across participants). The experiment consisted of 156 trials, in addition to 5 practice trials. Task condition (pro- and anti-saccade) and stimulus location (left and right) were randomly interleaved.

2.4. Stimuli

We used the International Affective Digitized Sound (IADS) database system (Bradley and Lang, 2007), with a set of acoustic emotional stimuli including contents across a wide range of semantic categories (e.g., female sigh, baby cry), because the IADS has been validated and extensively used to induce emotional arousal and applied to pupil research (e.g., Partala and Surakka, 2003). Although the database provides values of emotional valence and arousal level for each stimulus, we used the arousal and valence values reported by each participant to minimize the effects of individual differences. 8 IADS clips were excluded due to short duration (<6000 ms; clips 104, 204, 275, 296, 365, 400, 698, 699). Auditory stimuli were presented through headphones. The volume level was fixed across participants, set to a level that was predetermined to be comfortable and clear for all subjects.

2.5. Data analysis

Aspects of analyses related to task-evoked pupillary responses were published previously, and the exclusion criteria were described in detail (Cherng et al., 2020). Here, we focused on the time-frequency analysis of pupil size. Similar to our previous research (Cherng et al., 2020), the baseline-corrected epoch from 100 ms to event onset (emotional sounds or saccade cue) was used to baseline-correct pupil size. In FFT analysis, we applied fast Fourier transform using 'fft' function in Matlab to baseline-corrected pupil size, in which zeros was padded at the end to increase the frequency resolution up to 0.1 Hz and averaged the amplitude spectrum across trials. The amplitude spectrum can be represented in a single dimension frequency spectrum. In HHT analysis, we first applied an enhanced empirical mode decomposition (referred to as masking EMD) (Deering and Kaiser, 2005; Juan et al., 2021; Nguyen et al., 2019; Tsai et al., 2016), that is, using masking signals to decompose the raw epoched pupil oscillation data into twelve intrinsic mode functions (IMFs). This masking EMD resolved the mode-mixing problem in the original EMD and decomposed the intrinsic nature of the raw signal adaptively into single modes. After decomposition, Hilbert transform was used in each IMF to gather the local energy and instantaneous frequency information to produce HHT. In general, the analysis flow was conducted as follows:

The original signal $x(t)$ was adaptively decomposed into several intrinsic mode functions IMFs by using masking EMD (Fig. 1B). That is, each successive EMD, which based on sifting process, generated an IMF with lower oscillations than the previous IMF and naturally retained the physical properties of the signal and expressed as follows:

$$x(t) = \sum_{j=1}^n IMF_j(t) + r_n$$

Then, the Hilbert transform applied to estimate instantaneous frequencies and amplitudes of the IMFs. Thus, the original data can be expressed as:

$$x(t) = \sum_{j=1}^n a_j(t) \cos \theta_j(t) + r_n$$

where $a_j(t)$ and $\theta_j(t)$ are the instantaneous amplitude and phase of j th IMF. Then the instantaneous frequency is defined as the time derivative of the phase function:

$$f(t) = \frac{d\theta(t)}{dt}$$

This step produced the time-frequency characteristics of the original signal. Next, we projected these instantaneous amplitude and frequency to time-frequency space to form the spectrogram (from 0 to 3 Hz) in which the y-axis represents frequency, and the x-axis shows time (Fig. 1C). We then computed the spectrogram for each trial and then averaged the spectrogram across trials for each condition. To interpret the data and avoid the edges effects, 1 s of data length at the beginning and end was not shown in the spectrogram.

In addition to amplitude measures, inter-trial phase coherence (ITPC) was used to calculate the phase of trials, which reflects the temporal and phase synchronization. This index allowed us to investigate the phase locking of the temporal pupil dynamics (Schwiedrzik and Sudmann, 2020). To calculate the ITPC, we used empirical mode decomposition (EMD), which is a data-driven approach that eliminates the need for arbitrary bandpass filter cut-offs (Huang et al., 1998). Moreover, EMD-based phase locking analysis provides improved time resolution of synchrony compared to the bandpass filter (Sweeney-Reed and Nasuto, 2007). That is, the instantaneous phase was calculated by taking the angle of the analytic signal obtained using the Hilbert transform of each decomposed IMF. The equation of ITPC is as follows:

$$ITPC = \left| \frac{1}{N} \sum_{n=1}^N e^{j\phi(n,k)} \right|$$

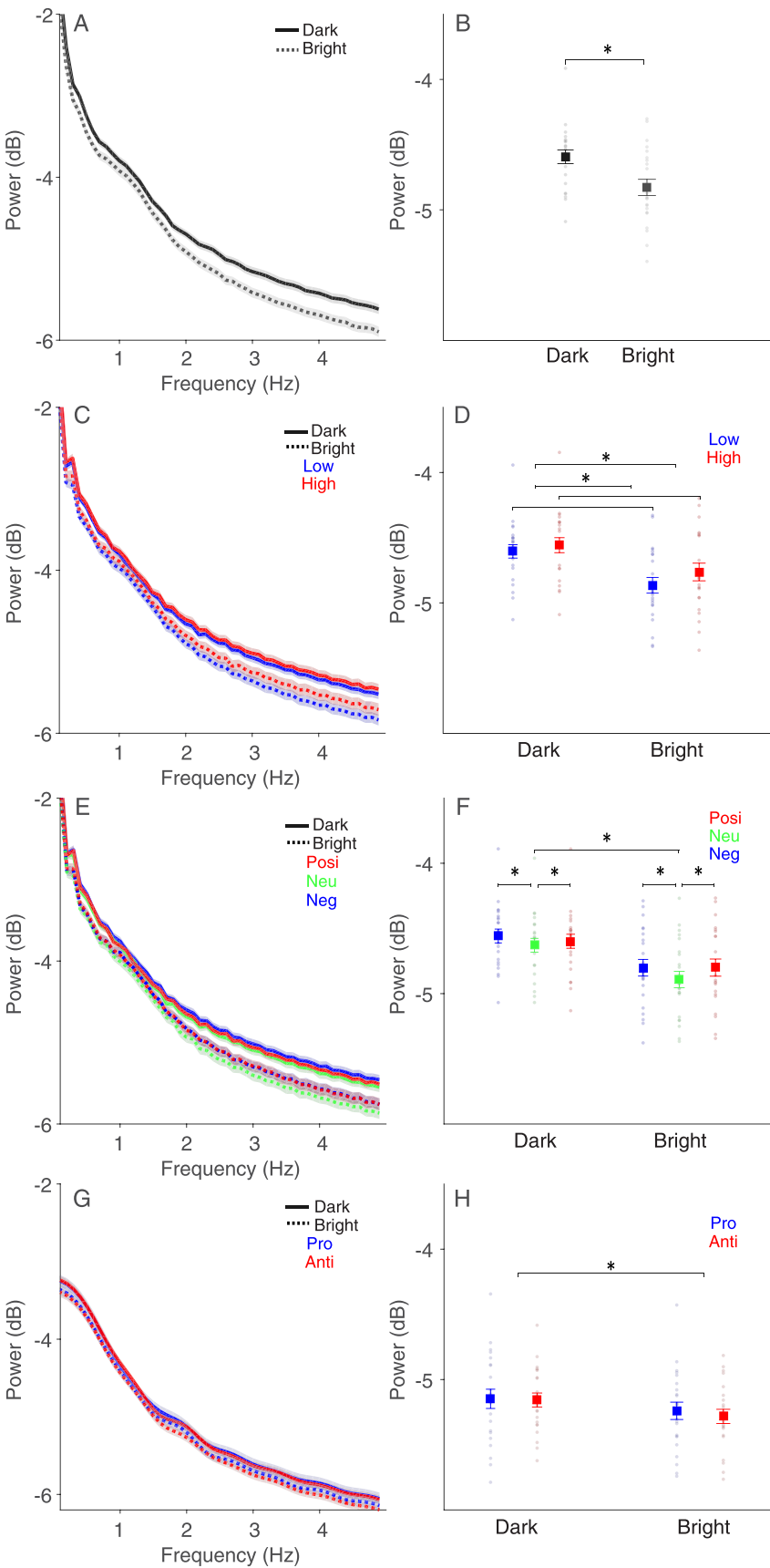


Fig. 2. Effect of background luminance, emotional arousal and valence, and saccade preparation on pupil oscillatory responses. Mean power spectrum of pupil oscillations on dark and bright backgrounds (A), and average Fourier power (N = 23) (B). Mean power spectrum of pupil oscillations during the periods emotional sounds in different arousal conditions with two backgrounds (C), and average Fourier power (N = 23) (D). Mean power spectrum of pupil oscillations during the periods emotional sounds in different valence conditions with two backgrounds (E), and average Fourier power (N = 23) (F). Mean power spectrum of pupil oscillations during the instructed fixation cue period in different saccade preparation conditions with two backgrounds (G), and average Fourier power (N = 23) (H). In A,C,E and G, the shaded colored regions surrounding the mean response curves represent the $\pm$ standard error range (across participants) for different conditions. In B,D,F and H, the squares and error-bars represent the mean values $\pm$ standard error across participants. The small-dots represent mean value for each participant. * Indicates differences are statistically significant.

where N is the number of trials and $\phi_{(n,k)}$ (rad.) is the instantaneous phase of the n^{th} trial and k^{th} IMF. Values of $ITPC$ were varied from 0 to 1 in which a value of 0 indicates a completely unalignment and 1 is identical phase angle across trials.

A two-way repeated-measure ANOVA was used to examine effects of background luminance (dark and bright) and emotional sounds (arousal or valence) on the pupil oscillatory response and Bonferroni-corrected t -test was used for the planned comparisons, except where indicated. We also examined effects of background luminance (dark and bright) and saccade preparation (pro- and anti-saccade) on the pupillary response. A two-tailed student t -test was performed to compare the differences between the two conditions. Effect sizes (partial eta squared or Cohen's d), where appropriate, were also reported. Statistical tests were performed using JASP Team (2019).

3. Results

3.1. Effects on pupillary oscillation using FFT

To understand the influence of global luminance level on pupillary oscillatory responses, we first used FFT to examine power spectral density in different background luminance levels during the period of emotional sounds (6000 ms) and saccade preparation cues (1400 ms). As shown in Fig. 2A, higher power was obtained with dark background (Fig. 2B, $t(22) = 6.76$, $p < 0.001$, $d = 1.41$). These results were consistent with previous studies showing higher power in the lower background luminance level (Murata and Iwase, 2002; Nakayama and Shimizu, 2004; Peysakhovich et al., 2015). To investigate whether pupillary oscillatory responses during the emotional sound presentation were modulated by affective processing, following our previous research (Cheng et al., 2020), we separated trials according to subjective emotional arousal or valence ratings and examined pupil size during 6 s of the emotional sound presentation. In emotional arousal analysis, we separated trials into two arousal categories (high: arousal > the median arousal value of a given subject; low: arousal < the median arousal value of a given subject) in order to have an approximately equal number of trials across conditions and subjects. As shown in Fig. 2C, results showed that higher power density observed in the high arousal condition than in the low arousal condition in both background luminance conditions (Fig. 2D: arousal main effects: $F(1,22) = 25.069$, $p < 0.001$, $\eta_p^2 = 0.533$; background luminance main effects: $F(1,22) = 38.812$, $p < 0.001$, $\eta_p^2 = 0.638$). Arousal effects were more pronounced with bright background (interaction: $F(1,22) = 5.13$, $p = 0.034$, $\eta_p^2 = 0.189$), though the simple main effects were significant in both background conditions ($p < 0.05$). We also explored the modulation of emotional valence on pupillary oscillatory responses by separating trials into three valence categories (negative: valence < 5; neutral: valence = 5; positive: valence > 5) according to the subjective valence ratings. As displayed in Fig. 2E, power spectral density was higher in the negative and positive valence conditions than in the neutral valence condition (Fig. 2F: $F(1,22) = 19.754$, $p < 0.001$, $\eta_p^2 = 0.473$), in addition to the global luminance modulation ($F(1,22) = 48.460$, $p < 0.001$, $\eta_p^2 = 0.688$). The interaction was not significant ($F(1,22) = 2.453$, $p = 0.098$, $\eta_p^2 = 0.1$). Notably, no differences were observed between the positive and negative sounds (all p s > 0.5). Finally, pupillary oscillatory responses during saccade preparation were also examined (1.4 s), and no significant differences in pupil oscillatory responses were obtained between pro- and anti-saccade preparation (Fig. 2G–H, saccade preparation: $F(1,22) = 1.184$, $p = 0.288$, $\eta_p^2 = 0.051$; interaction: $F(1,22) = 1.616$, $p = 0.217$, $\eta_p^2 = 0.068$), except for background luminance effects ($F(1,22) = 9.387$, $p = 0.006$, $\eta_p^2 = 0.299$). In summary, these results suggested that global luminance level modulated power spectral density in pupil oscillatory responses, and more importantly, emotional arousal also significantly modulated pupil oscillation, with higher power density observed in trials with emotional arousing stimuli.

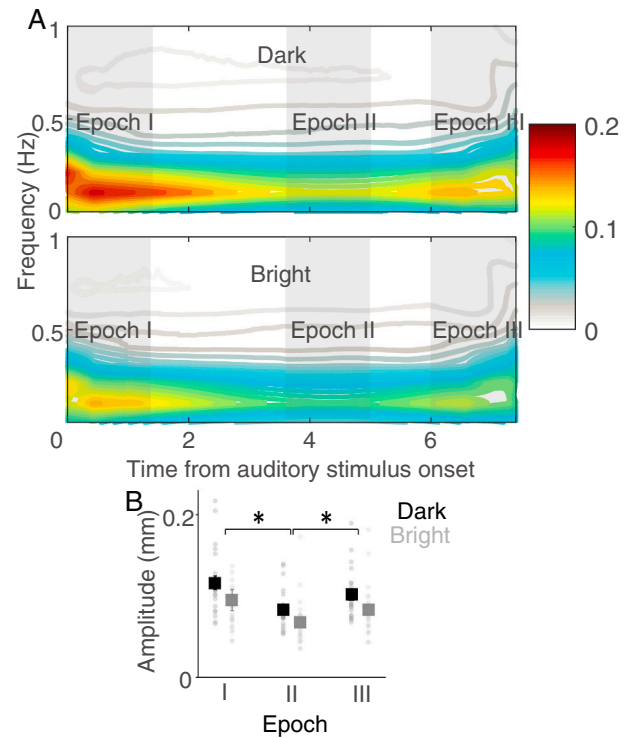


Fig. 3. Effect of time and frequency on pupil oscillatory responses. Heatmap density for amplitude spectrum dynamics after emotional sound onset on dark and bright backgrounds (A), and average power density across three time epochs (B). In A, the pupil epoch is shaded in gray. In B, the squares and error-bars represent the mean values $\pm$ standard error across participants. The small-dots represent mean value for each participant. * Indicates differences are statistically significant.

3.2. Time-frequency modulation on pupil size using HHT

To explore the time-frequency modulation on pupil size, HHT was used to analyze pupil data (combined all IMFs here). As displayed in Fig. 3A with the two background luminance conditions, pupillary oscillatory responses were dynamically changed across times, with higher power densities observed at the beginning and at the end period of the trial. To quantify this pattern of the results, we selected three time windows with the same duration (Epoch I: 1–1400 ms; Epoch II: 3600–5000 ms; Epoch III: 6000–7400 ms), frequency ranging from 0.1 Hz to 0.3 Hz with a higher power density. Results showed that power density was significantly different among these three time epochs (Fig. 3B; time window main effects: $F(1,22) = 23.846$, $p < 0.001$, $\eta_p^2 = 0.520$; background: $F(1,22) = 12.104$, $p = 0.002$, $\eta_p^2 = 0.355$; interaction: $F(1,22) = 0.558$, $p = 0.468$, $\eta_p^2 = 0.025$), with the lowest power in the middle time epoch. Overall, these results suggest that pupil oscillatory responses changed temporally, and HHT is thus useful to investigate the temporal modulation on pupil oscillations.

As shown in Fig. 1B, pupillary responses were particularly modulated by higher IMFs (i.e., lower oscillatory frequencies). The calculated amplitude using Hilbert transform on all IMFs (trials from different conditions were collapsed) also confirmed that lower frequency oscillations obtained with higher IMFs (Fig. 4), and pupil oscillations in these IMFs were mainly below 1 Hz, which is consistent with previous studies (Villalobos-Castaldi et al., 2016, 2017). The following analyses thus focused on the results of IMF 8–12.

3.3. Emotional arousal modulation on pupil oscillatory responses using HHT

To explore the time-frequency modulation of emotional arousal on

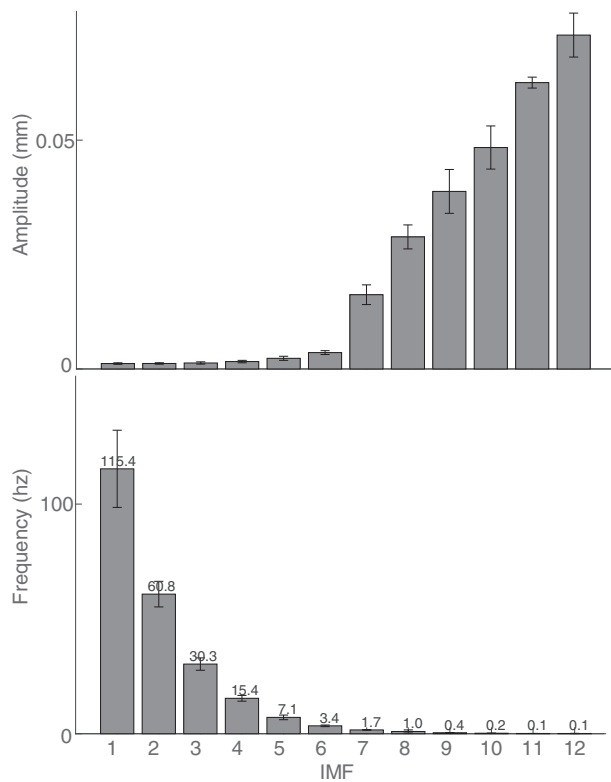


Fig. 4. HHT analysis for each IMF. The average amplitude and frequency in each IMF. The error-bar represents $\pm$ standard error across participants.

pupil size using HHT, following previous analysis, trials were separated into high and low emotional arousal categories. As mentioned, we focused here on IMF 8 to 12, and examined their power density dynamics and ITPC. As shown in Fig. 5A and B for the dark and bright condition, the amplitude spectrum changed dynamically across times in each IMF, and although not identical, these dynamics were similar in different background luminance conditions. Moreover, emotional arousal effects in each amplitude spectrum were also changed temporally. To quantify emotional arousal effects, the epoch from 3000 to 6000 ms of post-stimulus onset was used (Fig. 5C) because emotional arousal affects task-evoked pupillary responses in this epoch (Cherng et al., 2020). Results showed larger power density in the dark, compared to the bright condition in IMF 9, 10, and 12 (Fig. 5C, IMF 9: $F(1,22) = 7.144$, $p = 0.014$, $\eta_p^2 = 0.245$; IMF 10: $F(1,22) = 18.317$, $p < 0.001$, $\eta_p^2 = 0.454$; IMF 12: $F(1,22) = 20.611$, $p < 0.001$, $\eta_p^2 = 0.484$). Emotional arousal significantly affected pupillary responses in IMF 10 and 12, with a higher power in the high arousal condition (IMF 10: $F(1,22) = 9.746$, $p = 0.005$, $\eta_p^2 = 0.307$; IMF 12: $F(1,22) = 12.323$, $p = 0.002$, $\eta_p^2 = 0.359$). All other effects were not significant (all p s > 0.2). As shown in Fig. 6 for ITPC analysis, the synchronization of pupil oscillation changed across times in different IMFs (Fig. 6A and B), and the same epoch was used to quantify emotional arousal effects (Fig. 6C). Emotional arousal main effects and interaction were observed in IMF 11, showing higher coherence particularly in the high arousal condition (arousal: $F(1,22) = 7.394$, $p = 0.013$, $\eta_p^2 = 0.252$; interaction: $F(1,22) = 4.855$, $p = 0.038$, $\eta_p^2 = 0.181$). Moreover, in IMF 12, significantly higher coherence was obtained in the dark background ($F(1,22) = 9.316$, $p = 0.006$, $\eta_p^2 = 0.297$) and in the high emotional arousal condition ($F(1,22) = 4.891$, $p = 0.038$, $\eta_p^2 = 0.182$). Other effects were not significant (all p s > 0.3). Overall, these results suggested that time and frequency modulations on pupil size changed indeed dynamically, and higher power and ITPC

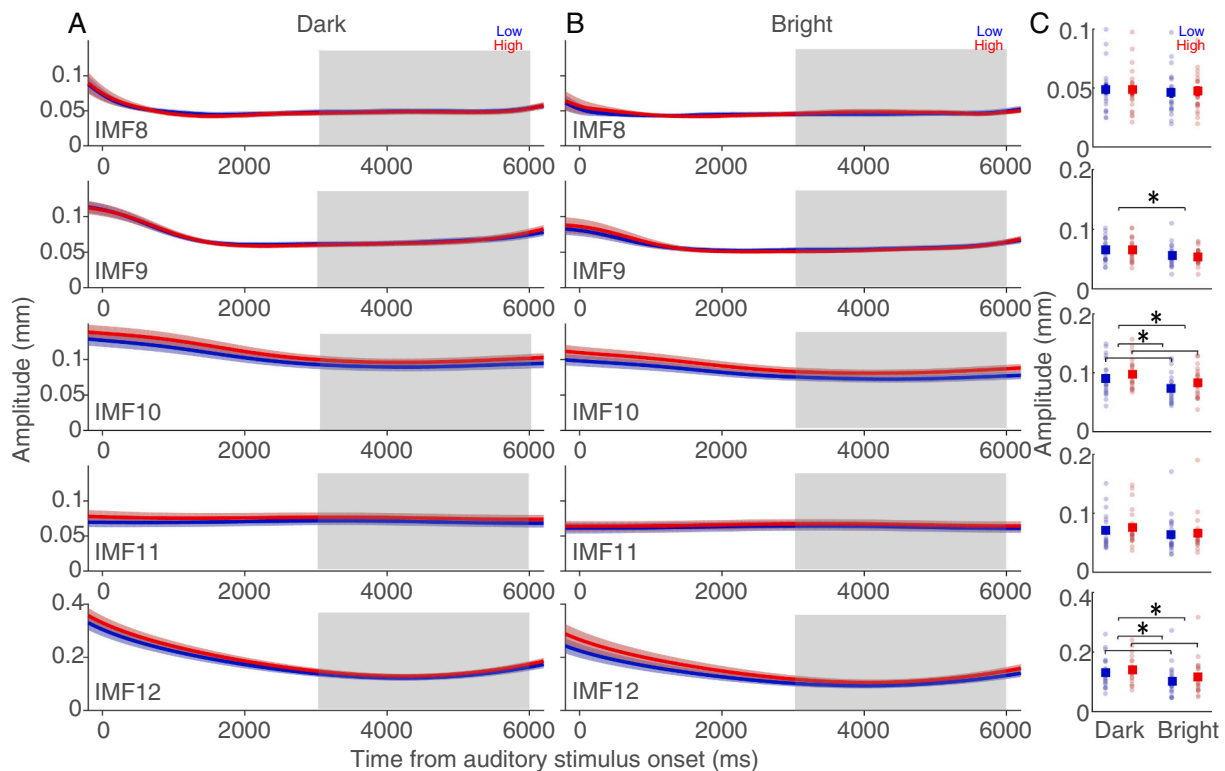


Fig. 5. Effect of time and frequency on pupil oscillatory responses associated with emotional arousal. Amplitude spectrum dynamics following emotional sounds in different arousal conditions in IMF 8 to 12 on dark (A) and, bright (B) backgrounds ($n = 23$). Average power density (3000 to 6000 ms) of pupil oscillatory responses shown for different arousal conditions with dark and bright backgrounds. In A and B, the shaded colored regions surrounding the mean response curves represent the $\pm$ standard error range (across participants) for different conditions. The pupil epoch is shaded in gray. In C, the squares and error-bars represent the mean values $\pm$ standard error across participants. The small-dots represent mean value for each participant. * Indicates differences are statistically significant.

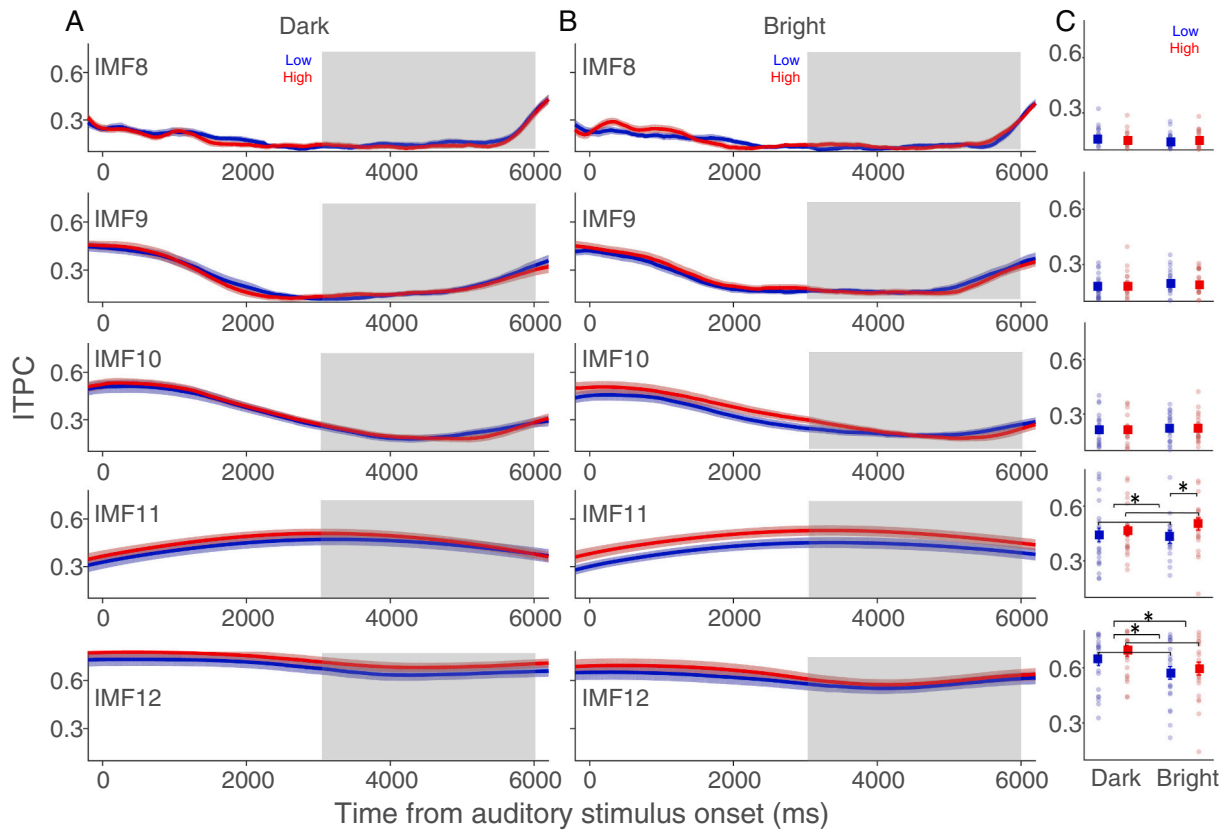


Fig. 6. Effect of time and frequency on pupil oscillation ITPC associated with emotional arousal. ITPC dynamics following emotional sounds in different arousal conditions in IMF 8 to 12 on dark (A) and, bright (B) backgrounds ($n = 23$). Average ITPC (3000 to 6000 ms) of pupil oscillation shown for different arousal conditions with dark and bright backgrounds. In A and B, the shaded colored regions surrounding the mean response curves represent the $\pm$ standard error range (across participants) for different conditions. The pupil epoch is shaded in gray. In C, the squares and error-bars represent the mean values $\pm$ standard error across participants. The small-dots represent mean value for each participant. * Indicates differences are statistically significant.

were obtained in the high emotional arousal condition, particularly in higher IMFs.

3.4. Saccade preparation modulation on pupil oscillatory responses using HHT

To examine pupil time-frequency characteristics during saccade preparation, trials were separated according to pro- or anti-saccade preparation condition. Similar to emotional arousal results, spectral amplitude constantly changed temporally in each IMF in the two background luminance conditions (Fig. 7A and B). To quantify saccade preparation effects (Fig. 7C), the epoch from 900 to 1400 ms of post-cue onset was used because saccade preparation effects on task-evoked pupillary responses in this epoch were particularly pronounced (Cherng et al., 2020; Wang et al., 2015). Saccade preparation significantly modulated spectral amplitude in all IMFs, with higher amplitude during pro-saccade preparation, compared to anti-saccade preparation (Fig. 7C: IMF 8: $F(1,22) = 11.268$, $p = 0.003$, $\eta_p^2 = 0.339$; IMF 9: $F(1,22) = 6.4$, $p = 0.019$, $\eta_p^2 = 0.225$; IMF 10: $F(1,22) = 6.521$, $p = 0.018$, $\eta_p^2 = 0.229$; IMF 11: $F(1,22) = 6.713$, $p = 0.017$, $\eta_p^2 = 0.234$; IMF 12: $F(1,22) = 20.917$, $p < 0.001$, $\eta_p^2 = 0.474$). Importantly, these saccade preparation modulations were statistically pronounced with bright background (all IMF simple main effects: $p < 0.05$), not with dark background (all $p > 0.08$), that is consistent with previous research showing a more reliable cognitive modulation on pupil size with bright background (Steinhauer et al., 2004). Moreover, consistently, larger power density with dark background were observed with IMF 9, 10, and 12 (IMF 9: $F(1,22) = 4.9$, $p = 0.038$, $\eta_p^2 = 0.182$; IMF 10: $F(1,22) = 5.313$, $p = 0.031$, $\eta_p^2 = 0.195$; IMF 12: $F(1,22) = 10.507$, $p = 0.004$, $\eta_p^2 = 0.323$).

In ITPC analysis, similar dynamics were observed between the two background conditions (Fig. 8A and B). The same epoch was used to quantify saccade preparation effects (Fig. 8C), showing saccade preparation significantly modulated ITPC in all IMFs, except IMF 12, though the direction of the modulation was different (IMF 8: $F(1,22) = 6.716$, $p = 0.017$, $\eta_p^2 = 0.234$; IMF 9: $F(1,22) = 29.392$, $p < 0.001$, $\eta_p^2 = 0.572$; IMF 10: $F(1,22) = 23.108$, $p < 0.001$, $\eta_p^2 = 0.512$; IMF 11: $F(1,22) = 6.713$, $p = 0.017$, $\eta_p^2 = 0.234$). Moreover, higher coherence was obtained with dark background in IMF 12 ($F(1,22) = 11.446$, $p = 0.003$, $\eta_p^2 = 0.342$). All other effects were not significant (all p s > 0.2). These results indeed reveal the dynamics between the frequency and temporal modulations on pupillary responses.

4. Discussion

The current study examined the time-frequency effects on pupil size modulated by global luminance, arousal, and cognitive signals using Hilbert-Huang Transform. Consistent with previous research, higher spectral density derived from FFT was obtained with a darker background. More interestingly, higher spectral density was also observed with higher levels of emotional arousal. These suggested that arousal also modulated pupil oscillations. With HHT, we further explored the time-frequency modulations, showing that amplitude spectrum and ITPC in each IMF changed temporally. Although both arousal and saccade preparation modulated pupil oscillatory responses, the modulation pattern in different IMFs could be even opposite. Given pupil oscillatory responses were indeed modulated by both time and frequency, and HHT is useful to investigate the temporal and frequency modulation on pupil oscillations.

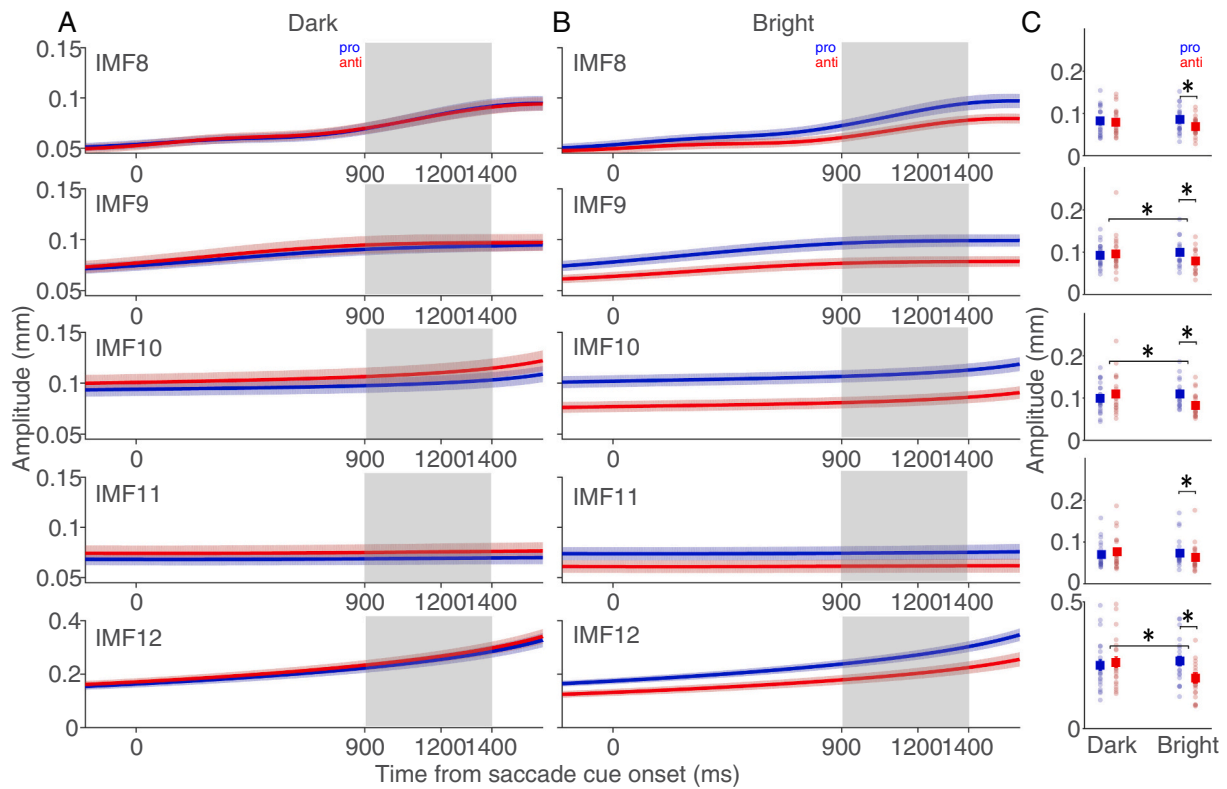


Fig. 7. Effect of time and frequency on pupil oscillatory responses related to saccade preparation. Amplitude spectrum dynamics following instructed fixation cue in different saccade preparation conditions in IMF 8 to 12 on dark (A) and, bright (B) backgrounds ($n = 23$). Average power density (900 to 1400 ms) of pupil oscillatory responses shown for different saccade preparation conditions with dark and bright backgrounds. In A and B, the shaded colored regions surrounding the mean response curves represent the $\pm$ standard error range (across participants) for different conditions. The pupil epoch is shaded in gray. In C, the squares and error-bars represent the mean values $\pm$ standard error across participants. The small-dots represent mean value for each participant. * Indicates differences are statistically significant.

Pupil oscillation modulated by global luminance, arousal, saccade preparation signals.

Spectral analysis using Fourier or Wavelet methods has been applied to pupil data, though these studies have mainly focused from an engineering perspective (Behrends et al., 2019; Lew et al., 2008; Lüdtke et al., 1998; Murata and Iwase, 2002; Naber et al., 2013; Nakayama and Shimizu, 2004; Nowak et al., 2008; Peysakhovich et al., 2015; Pomè et al., 2020; Reiner and Gelfeld, 2014; Ren et al., 2013; Schumann et al., 2020; Turnbull et al., 2017). Research has found higher power densities when the background luminance level is lower (Murata and Iwase, 2002; Nakayama and Shimizu, 2004; Peysakhovich et al., 2015). Consistently, here, using FFT we found higher power in the dark background condition, compared to the bright background condition. More importantly, power spectral density was also higher in trials with higher arousing sounds, suggesting that emotional arousal also modulated pupil oscillatory responses, in addition to its modulation on task-evoked pupillary responses. Although more errors occurred in the anti-saccade condition (more difficult) than in the pro-saccade condition (Cherng et al., 2020), there were no significant differences in spectral power density between the two saccade preparation conditions. Although these results were different from previous studies, showing higher power densities in the more difficult condition (Murata and Iwase, 2002; Nakayama and Shimizu, 2004; Peysakhovich et al., 2015), it is important to note that different effects are also observed (Murata and Iwase, 2002; Reiner and Gelfeld, 2014). Moreover, this inconsistency could be explained by differences in the temporal and frequency modulations on pupil oscillations that are difficult to reveal using the conventional spectral analytical methods. Overall, our results were consistent with the literature, and moreover, we demonstrated the modulation of emotional arousal on pupil oscillations.

HHT was used to further investigate the time-frequency modulation on pupil size. Previous research has used this method to analyze the spontaneous pupil oscillation (Villalobos-Castaldi et al., 2016, 2017), showing that the spectral density of the spontaneous pupil oscillation is concentrated at the low frequency range (mostly below 1 Hz), and these spontaneous oscillations vary over time. Consistently, here we found that pupil oscillations were significantly different at different points in time, with lowest amplitude in the mid period of the trial. Moreover, stronger modulations were obtained on higher IMFs (8–12), with dominant frequencies often below 1 Hz. Analyses further found that the modulation of emotional arousal, background luminance, and saccade preparation signals on different IMFs and its according ITPC was not always similar. For example: larger power densities were observed in the high arousal condition in some IMFs but not others, and background luminance and saccade preparation effects on amplitude spectrum or ITPC were sometimes different in different IMFs.

These results are important because they suggest that pupil oscillations are indeed modulated by the temporal factor. Moreover, different frequency ranges of pupil oscillations may be more sensitive to particular underlying processing, resulting in different modulations. This could potentially explain inconsistent findings on the pupil task difficulty spectral modulation (Murata and Iwase, 2002; Nakayama and Shimizu, 2004; Peysakhovich et al., 2015; Reiner and Gelfeld, 2014), as different task-difficulty manipulations could particularly influence some pupil oscillation frequencies but not others and possibly more pronounced at different points in time. Collapsed signals from both frequency and temporal domain are insufficient to characterize full pupil oscillatory modulations by the task difficulty manipulation from different studies. Notably, arousal or cognitive processing is a broad concept, and neural mechanisms involved in various arousal or

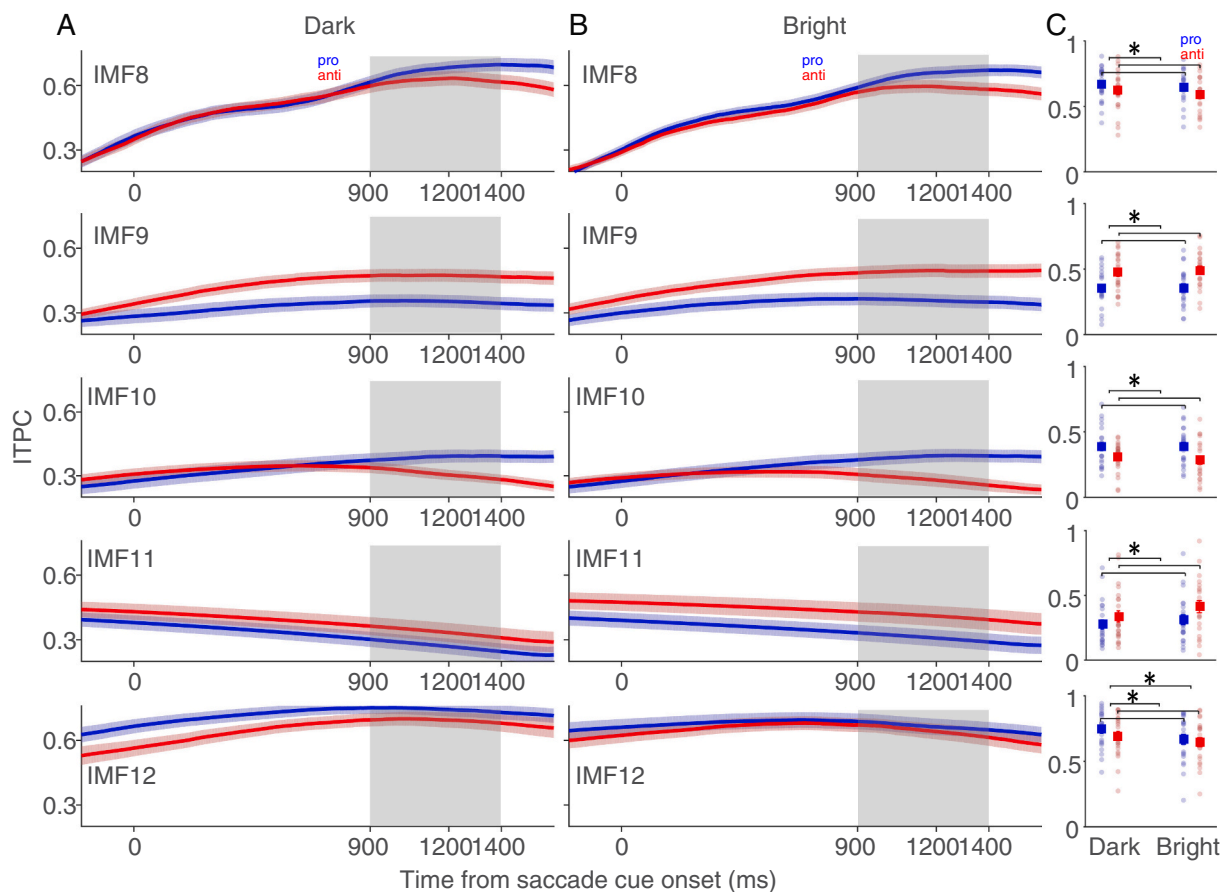


Fig. 8. Effect of time and frequency on pupil oscillation ITPC related to saccade preparation. ITPC dynamics following instructed fixation cue in different saccade preparation conditions in IMF 8 to 12 on dark (A) and, bright (B) backgrounds ($n = 23$). Average ITPC (900 to 1400 ms) of pupil oscillation shown for different saccade preparation conditions with dark and bright backgrounds. In A and B, the shaded colored regions surrounding the mean response curves represent the $\pm$ standard error range (across participants) for different conditions. The pupil epoch is shaded in gray. In C, the squares and error-bars represent the mean values $\pm$ standard error across participants. The small-dots represent mean value for each participant. * Indicates differences are statistically significant.

cognitive processes are certainly not identical, thus linking to the pupil control pathway differently. This could result in different time-frequency modulations on pupil oscillations. It is thus important to use HHT to examine various arousal and cognitive modulations on pupil oscillations to understand how pupil oscillatory responses are interactively modulated by time and frequency in the context of various pupil modulations.

5. Conclusion

Pupil size, as one of physiological signals, is likely constituted with non-linear and non-stationary waves. In the current study, through HHT, we systematically examined time-frequency characteristics of pupil oscillatory responses modulated by global luminance, arousal, and saccade preparation signals. We found that pupil oscillations were indeed modulated by temporal dimension. Effects of global luminance, arousal and saccade preparation on different IMFs were clearly different. Future research that uses HHT to understand pupil oscillations is needed to fully explore the time-frequency modulation on pupil size.

CRedit authorship contribution statement

C.W. designed research; C.W. and K.N. analyzed data and wrote the manuscript; all authors contributed to the study conception and

provided comments and edits on various drafts of the manuscript.

Declaration of competing interest

The authors report no conflicts of interest.

Data availability

Data will be made available on request.

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Data availability

Data are available upon reasonable request following the publication.

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